ALTINBAŞ UNIVERSITY
Faculty of Engineering and Natural Sciences
Department of Software Engineering

Graduation Project

SkillStream: Multi-Role Online Learning Platform

Progress Report: Week 1 and Week 2

Course	Graduation Project
Institution	Altınbaş University
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Reporting Period	Week 1 – Week 2
Date	October 2026

Part I: Week 1 – Project Definition

1.1 Project Title

SkillStream: Multi-Role Online Learning Platform.

1.2 Problem Statement

Online learning platforms frequently encounter difficulties in organising and distributing permissions among administrators, instructors, and learners. When responsibilities are not clearly separated, instructors depend on administrative staff for routine content updates, administrators lose visibility over the quality of published material, and students are presented with fragmented interfaces that hinder course discovery and consumption.

Instructors therefore require a straightforward means of publishing and maintaining their own course content, whereas students require an intuitive environment in which they can browse the course catalogue, enrol in courses, and watch lessons without interruption. SkillStream addresses this problem by providing a single web-based platform built around a clearly defined role-based access model comprising three roles: Admin, Teacher, and Student.

1.3 Objectives

The principal objectives of the project are as follows:

- To design and implement a role-based access control model that clearly separates the privileges of Admin, Teacher, and Student users.
- To enable teachers to create, update, and publish courses composed of structured video lessons.
- To allow students to browse a searchable course catalogue, enrol in courses, and track their learning progress.
- To provide administrators with tools for managing user accounts and moderating published courses.
- To deliver a secure and responsive system that satisfies the defined non-functional requirements for performance and data protection.
- To apply a disciplined software engineering process, including requirements engineering, version control, and iterative delivery, within a nine-week schedule.

1.4 Academic and Technical References

[1] Sommerville, I. (2016). Software Engineering (10th ed.). Pearson.

[2] Pressman, R. S., & Maxim, B. R. (2020). Software Engineering: A Practitioner's Approach (9th ed.). McGraw-Hill Education.

[3] ISO/IEC/IEEE 29148:2018. Systems and Software Engineering – Life Cycle Processes – Requirements Engineering. International Organization for Standardization.

[4] Sandhu, R. S., Coyne, E. J., Feinstein, H. L., & Youman, C. E. (1996). Role-based access control models. IEEE Computer, 29(2), 38–47.

[5] Wiegers, K., & Beatty, J. (2013). Software Requirements (3rd ed.). Microsoft Press.

[6] OWASP Foundation. Password Storage Cheat Sheet. OWASP Cheat Sheet Series. https://cheatsheetseries.owasp.org/cheatsheets/Password_Storage_Cheat_Sheet.html

[7] Google. Web Vitals. web.dev. https://web.dev/articles/vitals

Part II: Week 2 – Requirements and Planning

2.1 Functional Requirements

ID	Requirement
FR-01	The system shall allow users to register and log in as Admin, Teacher, or Student.
FR-02	The system shall allow administrators to manage platform users and moderate courses.
FR-03	The system shall allow teachers to create, update, and publish course content and video lessons.
FR-04	The system shall allow students to browse the course catalog and enroll in available courses.
FR-05	The system shall allow enrolled students to watch video lessons and track their progress.

2.2 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	The system shall load course catalog pages in under 2 seconds under normal network conditions.
NFR-02	Security	User passwords must be securely hashed before being stored in the database.

2.3 Project Scope

In Scope

- Management of the three user roles (Admin, Teacher, Student) and their respective permissions.
- Course creation, editing, and publishing by teachers.
- Student registration, course enrolment, and viewing of course content.
- Tracking of lesson completion and overall course progress for enrolled students.

Out of Scope

- Online payment gateways and paid course transactions.
- Live (real-time) video streaming.

2.4 Constraints

- Time: the project must be completed and submitted within the academic semester (nine weeks).

- Resources: development is carried out by a small student team using free or open-source tools and services.
- Hosting: video content is limited to pre-recorded files within the storage limits of the selected hosting environment.

2.5 Assumptions

- Users have access to a modern web browser and a stable internet connection.
- Teachers provide their own pre-recorded video lessons in commonly supported formats.
- Each user account is associated with exactly one role at any given time.
- Administrators are trusted users responsible for approving or removing inappropriate content.

2.6 Nine-Week Project Milestones

Week	Milestone	Main Deliverables	Status
1	Project definition	Title, problem statement, objectives, references	Completed
2	Requirements and planning	FR/NFR, scope, constraints, assumptions, milestones	Completed
3	System design	Use case diagram, ER diagram, system architecture	Planned
4	UI/UX design	Wireframes and page flow for each role	Planned
5	Authentication and roles	Registration, login, role-based access (FR-01)	Planned
6	Course management	Teacher course creation and admin moderation (FR-02, FR-03)	Planned
7	Learning features	Catalogue, enrolment, video lessons, progress (FR-04, FR-05)	Planned
8	Testing and refinement	Functional, performance and security testing (NFR-01, NFR-02)	Planned
9	Final delivery	Deployment, final report and presentation	Planned

Part III: Project Repository

The source code, documentation, and task tracking for the project are maintained on GitHub:

Item	Link
Repository URL	https://github.com/munthermohammed064-pixel/skillstream-learning-platform
Issues Page	https://github.com/munthermohammed064-pixel/skillstream-learning-platform/issues
Commit History	https://github.com/munthermohammed064-pixel/skillstream-learning-platform/commits/main